Assignment 2:

A) Setup

```
nano lab_2.txt
```

```
sed -i 's/^/STUDENT_ID/' lab_2.txt
```

B) grep

1. Extract all lines from lab_2.txt that contain the word "INFO." How many lines did you find?

```
grep "INFO" lab_2.txt
```

```
STUDENT_ID 2024-09-23 12:00:00 INFO Process started
STUDENT_ID 2024-09-23 12:00:45 INFO Log file initialized: /var/log/lab_process.log
STUDENT_ID 2024-09-23 12:01:30 INFO Setting default values for parameters
STUDENT_ID 2024-09-23 12:02:00 INFO Loading user preferences from
STUDENT_ID 2024-09-23 12:02:30 INFO Reading configuration files from /etc/config.yaml
STUDENT_ID 2024-09-23 12:03:00 INFO Configuration loaded successfully
STUDENT_ID 2024-09-23 12:04:00 INFO Connecting to the database at db.example.com
STUDENT_ID 2024-09-23 12:04:30 INFO Database connection established successfully
STUDENT_ID 2024-09-23 12:05:45 INFO User requested data analysis on input files
STUDENT_ID 2024-09-23 12:07:00 INFO Checking if user has read access to /data/input/
STUDENT_ID 2024-09-23 12:08:30 INFO Suggesting user request access to restricted
STUDENT_ID 2024-09-23 12:09:00 INFO Executing fallback data retrieval method
STUDENT_ID 2024-09-23 12:10:30 INFO Monitoring memory usage patterns
STUDENT_ID 2024-09-23 12:11:30 INFO Attempting to optimize memory allocation
STUDENT_ID 2024-09-23 12:12:30 INFO Logging error to error log:
STUDENT_ID 2024-09-23 12:13:00 INFO Searching for alternative file paths
STUDENT_ID 2024-09-23 12:13:30 INFO Checking /data/backup/ for necessary files
STUDENT_ID 2024-09-23 12:14:00 INFO File not found in alternative paths; user notified
STUDENT_ID 2024-09-23 12:14:30 INFO Preparing to execute search commands
STUDENT_ID 2024-09-23 12:15:00 INFO Executing command: grep -i "test" file.txt
STUDENT_ID 2024-09-23 12:16:00 INFO Command executed with parameters: grep -i "test"
STUDENT_ID 2024-09-23 12:17:00 INFO Capturing output lines that match the search criteria
STUDENT_ID 2024-09-23 12:18:00 INFO Results of grep command:
STUDENT_ID 2024-09-23 12:18:30 INFO Matching line: "This is a test line in the file."
STUDENT_ID 2024-09-23 12:19:00 INFO Additional matching line: "Another test case found."
STUDENT_ID 2024-09-23 12:20:00 INFO Command completed successfully
STUDENT_ID 2024-09-23 12:21:00 INFO Aggregating results for reporting
STUDENT_ID 2024-09-23 12:21:30 INFO Preparing summary report for the user
STUDENT_ID 2024-09-23 12:23:00 INFO Total matches found: 2
STUDENT_ID 2024-09-23 12:31:00 INFO Checking file integrity
STUDENT_ID 2024-09-23 12:31:30 INFO Performing checksum verification on input files
STUDENT_ID 2024-09-23 12:32:00 INFO File integrity check completed without issues
STUDENT_ID 2024-09-23 12:32:30 INFO Logging integrity check results
STUDENT_ID 2024-09-23 12:35:00 INFO Line count: 100
STUDENT_ID 2024-09-23 12:36:00 INFO Calculating processing time
STUDENT_ID 2024-09-23 12:37:30 INFO Error details: Insufficient memory to complete
STUDENT_ID 2024-09-23 12:38:00 INFO Attempting to free up memory
STUDENT_ID 2024-09-23 12:38:30 INFO Running garbage collection to reclaim memory
STUDENT_ID 2024-09-23 12:39:00 INFO Cleanup process initiated
STUDENT_ID 2024-09-23 12:40:00 INFO Closing unused resources
STUDENT_ID 2024-09-23 12:41:00 INFO Cleanup completed successfully
STUDENT_ID 2024-09-23 12:42:00 INFO Retrying previous operation
STUDENT_ID 2024-09-23 12:42:30 INFO Operation resumed successfully after cleanup
STUDENT_ID 2024-09-23 12:43:00 INFO Finalizing report generation
STUDENT_ID 2024-09-23 12:44:00 INFO Operation completed successfully after cleanup
STUDENT_ID 2024-09-23 12:45:00 INFO End of log
```

```
grep -c "INFO" lab_2.txt
```

```
46
```

Found 46 lines.

2. Find and display all lines that include the term "ERROR" case insensitive.

```
grep -i "ERROR" lab_2.txt
```

```
STUDENT_ID 2024-09-23 12:12:00 ERROR Could not find file /path/to/file
STUDENT_ID 2024-09-23 12:12:30 INFO Logging error to error log:
STUDENT_ID /var/log/lab_process_error.log
STUDENT_ID 2024-09-23 12:37:00 ERROR An unexpected error occurred
STUDENT_ID 2024-09-23 12:37:30 INFO Error details: Insufficient memory to complete
```

3. Use grep to count how many times the word "user" appears in lab_2.txt. What are the matching lines?

```
grep -ow "user" lab_2.txt | wc -l
```

```
7
```

```
grep -w "user" lab_2.txt
```

```
STUDENT_ID 2024-09-23 12:02:00 INFO Loading user preferences from
STUDENT_ID /home/user/.config/preferences.yaml
STUDENT_ID 2024-09-23 12:06:00 DEBUG Validating user permissions for data access
STUDENT_ID 2024-09-23 12:07:00 INFO Checking if user has read access to /data/input/
STUDENT_ID 2024-09-23 12:08:30 INFO Suggesting user request access to restricted
STUDENT_ID 2024-09-23 12:14:00 INFO File not found in alternative paths; user notified
STUDENT_ID 2024-09-23 12:21:30 INFO Preparing summary report for the user
```

4. Search for lines that contain the word "memory" or “connection” and print those lines along with their line numbers.

```
grep -nE "memory|connection" lab_2.txt
```

```
12:STUDENT_ID 2024-09-23 12:04:30 INFO Database connection established successfully
23:STUDENT_ID 2024-09-23 12:10:30 INFO Monitoring memory usage patterns
24:STUDENT_ID 2024-09-23 12:11:30 INFO Attempting to optimize memory allocation
55:STUDENT_ID 2024-09-23 12:37:30 INFO Error details: Insufficient memory to complete
57:STUDENT_ID 2024-09-23 12:38:00 INFO Attempting to free up memory
58:STUDENT_ID 2024-09-23 12:38:30 INFO Running garbage collection to reclaim memory
```

5. Extract all lines containing "file" and display the total number of matches.

```
grep "file" lab 2.txt
STUDENT_ID 2024-09-23 12:00:15 DEBUG Checking for existing log files to prevent overwriting
STUDENT_ID 2024-09-23 12:00:45 INFO Log file initialized: /var/log/lab_process.log
STUDENT_ID 2024-09-23 12:02:30 INFO Reading configuration files from /etc/config.yaml
STUDENT_ID 2024-09-23 12:05:45 INFO User requested data analysis on input files
STUDENT_ID 2024-09-23 12:12:00 ERROR Could not find file /path/to/file
STUDENT_ID 2024-09-23 12:13:00 INFO Searching for alternative file paths
STUDENT_ID 2024-09-23 12:13:30 INFO Checking /data/backup/ for necessary files
STUDENT_ID 2024-09-23 12:15:00 INFO Executing command: grep -i "test" file.txt
STUDENT_ID /data/input/file.txt
STUDENT_ID 2024-09-23 12:18:30 INFO Matching line: "This is a test line in the file."
STUDENT_ID 2024-09-23 12:30:00 DEBUG Searching for patterns in files
STUDENT_ID 2024-09-23 12:31:00 INFO Checking file integrity
STUDENT_ID 2024-09-23 12:31:30 INFO Performing checksum verification on input files
```

```
grep -c "file" lab_2.txt
13
```

Total Number of Matches is 13.

C) sed

1. Replace all occurrences of the word "INFO" in lab_2.txt with "INFORMATION." Save the output to a new file called updated_log.txt.

```
sed 's/INFO/INFORMATION/g' lab_2.txt > updated_log.txt
```

2. Extract only the timestamp and message from lines that contain "WARNING" and write the output to a file named warnings.txt.

```
sed -n '/WARNING/s/^\([^ ]* [^ ]*\) WARNING /\1 /p' lab 2.txt > warnings.txt
```

3. Remove all lines that start with "DEBUG" from lab_2.txt and save the cleaned output to cleaned_log.txt.

```
sed '/DEBUG/d' lab 2.txt > cleaned log.txt
```

4. Change all occurrences of "user" to "student" in lab_2.txt. Save the modified file as modified_log.txt.

```
sed 's/user/student/gI' lab 2.txt > modified log.txt
```

5. Use sed to insert the text "Log Entry:" at the beginning of each line in lab_2.txt and save it to labeled_log.txt.

```
sed 's/^/Log Entry: /' lab_2.txt > labeled_log.txt
```

6. Delete all lines that do not contain the word "successfully" and save the output to success_log.txt.

```
sed '/successfully/!d' lab_2.txt > success_log.txt
```

D) awk

1. Use awk to print the timestamps and messages from lab_2.txt, but only for lines where the severity level is "ERROR."

```
awk '$3=="ERROR" {$3=""; print}' lab_2.txt
```

2. Count the total number of lines in lab_2.txt using awk and print the result.

```
awk 'END {print "Total lines:", NR}' lab_2.txt
 66
```

3. Display only the messages from lines that contain "Executing command" and the corresponding timestamps.

```
awk '/Executing command/ {printf "%s %s ", $1, $2; for(i=4;i<=NF;i++) printf "
%s ", $i; print ""}' lab_2.txt
STUDENT ID 2024-09-23 INFO Executing command: grep -i "test" file.txt
```

4. Calculate and display the total number of "INFO" lines in lab_2.txt using awk.

```
awk '$3=="INFO" {count++} END {print count}' lab_2.txt
```

Blank, meaning 0 lines.

5. Print lines from lab_2.txt where it has “memory”, along with the timestamps.

```
awk '/memory/ {print $1, $2, $0}' lab 2.txt
```

```
STUDENT_ID 2024-09-23 STUDENT_ID 2024-09-23 12:10:30 INFO Monitoring memory usage patterns
STUDENT_ID 2024-09-23 STUDENT_ID 2024-09-23 12:11:30 INFO Attempting to optimize memory allocation
STUDENT_ID 2024-09-23 STUDENT_ID 2024-09-23 12:37:30 INFO Error details: Insufficient memory to complete
STUDENT_ID 2024-09-23 STUDENT_ID 2024-09-23 12:38:00 INFO Attempting to free up memory
STUDENT_ID 2024-09-23 STUDENT_ID 2024-09-23 12:38:30 INFO Running garbage collection to reclaim memory
```

6. Filter and print lines that mention "access" and display their line numbers.

```
awk '/access/ {print NR, $0}' lab 2.txt
```

```
15 STUDENT_ID 2024-09-23 12:06:00 DEBUG Validating user permissions for data access
16 STUDENT_ID 2024-09-23 12:07:00 INFO Checking if user has read access to /data/input/
17 STUDENT_ID 2024-09-23 12:08:00 WARNING User permissions insufficient for accessing
19 STUDENT_ID 2024-09-23 12:08:30 INFO Suggesting user request access to restricted
```

E) Tar

1. Create a parent directory with the structure: a. lab2/dir1/dir2/dir3 b. lab2/dir1/dir4

```
mkdir -p lab2/dir1/dir2/dir3 lab2/dir1/dir4
```

2. Move all files to lab2/dir1/dir2/dir3

```
mv lab 2.txt updated log.txt warnings.txt cleaned log.txt modified log.txt lab
eled_log.txt success_log.txt lab2/dir1/dir2/dir3/
```

3. Archive and compress the folder using tar

```
tar -czvf lab2.tar.gz lab2/
```

4. Print contents of archived folder

```
ubuntu@ip-172-31-20-99:~$ nano lab_2.txt
ubuntu@ip-172-31-20-99:~$ ls STUDENT_ID_assignment/
lesson1
ubuntu@ip-172-31-20-99:~$ sed -i 's/^/STUDENT_ID /' lab_2.txt
ubuntu@ip-172-31-20-99:~$ grep -c "INFO" lab_2.txt
46
ubuntu@ip-172-31-20-99:~$ grep -i "ERROR" lab_2.txt
STUDENT_ID 2024-09-23 12:12:00 ERROR Could not find file /path/to/file
STUDENT_ID 2024-09-23 12:12:30 INFO Logging error to error log:
STUDENT_ID /var/log/lab_process_error.log
STUDENT_ID 2024-09-23 12:37:00 ERROR An unexpected error occurred
STUDENT_ID 2024-09-23 12:37:30 INFO Error details: Insufficient memory to complete
ubuntu@ip-172-31-20-99:~$ grep -ow "user" lab_2.txt | wc -l
7
ubuntu@ip-172-31-20-99:~$ grep -w "user" lab_2.txt
STUDENT_ID 2024-09-23 12:02:00 INFO Loading user preferences from
STUDENT_ID /home/user/.config/preferences.yaml
STUDENT_ID 2024-09-23 12:06:00 DEBUG Validating user permissions for data access
STUDENT_ID 2024-09-23 12:07:00 INFO Checking if user has read access to /data/input/
STUDENT_ID 2024-09-23 12:08:30 INFO Suggesting user request access to restricted
STUDENT_ID 2024-09-23 12:14:00 INFO File not found in alternative paths; user notified
STUDENT_ID 2024-09-23 12:21:30 INFO Preparing summary report for the user
ubuntu@ip-172-31-20-99:~$ grep -nE "memory
> ^C
ubuntu@ip-172-31-20-99:~$ grep -nE "memory|connection" lab_2.txt
12:STUDENT_ID 2024-09-23 12:04:30 INFO Database connection established successfully
23:STUDENT_ID 2024-09-23 12:10:30 INFO Monitoring memory usage patterns
24:STUDENT_ID 2024-09-23 12:11:30 INFO Attempting to optimize memory allocation
55:STUDENT_ID 2024-09-23 12:37:30 INFO Error details: Insufficient memory to complete
57:STUDENT_ID 2024-09-23 12:38:00 INFO Attempting to free up memory
58:STUDENT_ID 2024-09-23 12:38:30 INFO Running garbage collection to reclaim memory
ubuntu@ip-172-31-20-99:~$ grep "file" lab_2.txt
```

```
STUDENT_ID 2024-09-23 12:00:15 DEBUG Checking for existing log files to prevent overwriting
STUDENT_ID 2024-09-23 12:00:45 INFO Log file initialized: /var/log/lab_process.log
STUDENT_ID 2024-09-23 12:02:30 INFO Reading configuration files from /etc/config.yaml
STUDENT_ID 2024-09-23 12:05:45 INFO User requested data analysis on input files
STUDENT_ID 2024-09-23 12:12:00 ERROR Could not find file /path/to/file
STUDENT_ID 2024-09-23 12:13:00 INFO Searching for alternative file paths
STUDENT_ID 2024-09-23 12:13:30 INFO Checking /data/backup/ for necessary files
STUDENT_ID 2024-09-23 12:15:00 INFO Executing command: grep -i "test" file.txt
STUDENT_ID /data/input/file.txt
STUDENT_ID 2024-09-23 12:18:30 INFO Matching line: "This is a test line in the file."
STUDENT_ID 2024-09-23 12:30:00 DEBUG Searching for patterns in files
STUDENT_ID 2024-09-23 12:31:00 INFO Checking file integrity
STUDENT_ID 2024-09-23 12:31:30 INFO Performing checksum verification on input files
ubuntu@ip-172-31-20-99:~$ grep -c "file" lab_2.txt
13
ubuntu@ip-172-31-20-99:~$ sed 's/INFO/INFORMATION/g' lab_2.txt > updated_log.txt
ubuntu@ip-172-31-20-99:~$ sed -n '/WARNING/s/^\([^ ]* [^ ]*\) WARNING /\1 /p' lab_2.txt > warnings.txt
ubuntu@ip-172-31-20-99:~$ sed '/DEBUG/d' lab_2.txt > cleaned_log.txt
ubuntu@ip-172-31-20-99:~$ sed 's/user/student/gI' lab_2.txt > modified_log.txt
ubuntu@ip-172-31-20-99:~$ sed 's/^/Log Entry: /' lab_2.txt > labeled_log.txt
ubuntu@ip-172-31-20-99:~$ sed '/successfully/!d' lab_2.txt > success_log.txt
ubuntu@ip-172-31-20-99:~$ awk '$3=="ERROR" {$3=""; print}' lab_2.txt
ubuntu@ip-172-31-20-99:~$ awk 'END {print "Total lines:", NR}' lab_2.txt
Total lines: 66
ubuntu@ip-172-31-20-99:~$ awk '/Executing command/ {printf "%s %s ", $1, $2; for(i=4;i<=NF;i++) printf "
%s ", $i; print ""}' lab_2.txt
STUDENT_ID 2024-09-23 INFO Executing command: grep -i "test" file.txt
ubuntu@ip-172-31-20-99:~$ awk '$3=="INFO" {count++} END {print count}' lab_2.txt

ubuntu@ip-172-31-20-99:~$ awk '/memory/ {print $1, $2, $0}' lab_2.txt
STUDENT_ID 2024-09-23 STUDENT_ID 2024-09-23 12:10:30 INFO Monitoring memory usage patterns
STUDENT_ID 2024-09-23 STUDENT_ID 2024-09-23 12:11:30 INFO Attempting to optimize memory allocation
STUDENT_ID 2024-09-23 STUDENT_ID 2024-09-23 12:37:30 INFO Error details: Insufficient memory to complete
STUDENT_ID 2024-09-23 STUDENT_ID 2024-09-23 12:38:00 INFO Attempting to free up memory
STUDENT_ID 2024-09-23 STUDENT_ID 2024-09-23 12:38:30 INFO Running garbage collection to reclaim memory
ubuntu@ip-172-31-20-99:~$ awk '/access/ {print NR, $0}' lab_2.txt
15 STUDENT_ID 2024-09-23 12:06:00 DEBUG Validating user permissions for data access
16 STUDENT_ID 2024-09-23 12:07:00 INFO Checking if user has read access to /data/input/
17 STUDENT_ID 2024-09-23 12:08:00 WARNING User permissions insufficient for accessing
19 STUDENT_ID 2024-09-23 12:08:30 INFO Suggesting user request access to restricted
ubuntu@ip-172-31-20-99:~$ mkdir -p lab2/dir1/dir2/dir3 lab2/dir1/dir4
ubuntu@ip-172-31-20-99:~$ mv lab_2.txt updated_log.txt warnings.txt cleaned_log.txt modified_log.txt lab
eled_log.txt success_log.txt lab2/dir1/dir2/dir3/
ubuntu@ip-172-31-20-99:~$ tar -czvf lab2.tar.gz lab2/
lab2/
lab2/dir1/
lab2/dir1/dir4/
```

```
lab2/dir1/dir2/
lab2/dir1/dir2/dir3/
lab2/dir1/dir2/dir3/warnings.txt
lab2/dir1/dir2/dir3/modified_log.txt
lab2/dir1/dir2/dir3/updated_log.txt
lab2/dir1/dir2/dir3/cleaned_log.txt
lab2/dir1/dir2/dir3/labeled_log.txt
lab2/dir1/dir2/dir3/lab_2.txt
lab2/dir1/dir2/dir3/success_log.txt
ubuntu@ip-172-31-20-99:~$ tar -tzvf lab2.tar.gz
drwxrwxr-x ubuntu/ubuntu     0 2026-10-14 22:57 lab2/
drwxrwxr-x ubuntu/ubuntu     0 2026-10-14 22:57 lab2/dir1/
drwxrwxr-x ubuntu/ubuntu     0 2026-10-14 22:57 lab2/dir1/dir4/
drwxrwxr-x ubuntu/ubuntu     0 2026-10-14 22:57 lab2/dir1/dir2/
drwxrwxr-x ubuntu/ubuntu     0 2026-10-14 22:58 lab2/dir1/dir2/dir3/
-rw-rw-r-- ubuntu/ubuntu     0 2026-10-14 22:49 lab2/dir1/dir2/dir3/warnings.txt
-rw-rw-r-- ubuntu/ubuntu  4620 2026-10-14 22:50 lab2/dir1/dir2/dir3/modified_log.txt
-rw-rw-r-- ubuntu/ubuntu  4912 2026-10-14 22:45 lab2/dir1/dir2/dir3/updated_log.txt
-rw-rw-r-- ubuntu/ubuntu  3838 2026-10-14 22:50 lab2/dir1/dir2/dir3/cleaned_log.txt
-rw-rw-r-- ubuntu/ubuntu  5316 2026-10-14 22:51 lab2/dir1/dir2/dir3/labeled_log.txt
-rw-rw-r-- ubuntu/ubuntu  4590 2026-10-14 22:35 lab2/dir1/dir2/dir3/lab_2.txt
-rw-rw-r-- ubuntu/ubuntu   449 2026-10-14 22:51 lab2/dir1/dir2/dir3/success_log.txt
```

5. Remove the original folder

```
rm -r lab2
```

6. Unarchive the tar.gz file

```
tar -xzvf lab2.tar.gzs
```

File History:

```
  247  nano lab_2.txt
  248  ls STUDENT_ID_assignment/
  249  sed -i 's/^/STUDENT_ID /' lab_2.txt
  250  grep -c "INFO" lab_2.txt
  251  grep -i "ERROR" lab_2.txt
  252  grep -ow "user" lab_2.txt | wc -l
  253  grep -w "user" lab_2.txt
  254  grep -nE "memory
  255  grep -nE "memory|connection" lab_2.txt
  256  grep "file" lab_2.txt
  257  grep -c "file" lab_2.txt
  258  sed 's/INFO/INFORMATION/g' lab_2.txt > updated_log.txt
  259  sed -n '/WARNING/s/^\([^ ]* [^ ]*\) WARNING /\1 /p' lab_2.txt > warnings.txt
  260  sed '/DEBUG/d' lab_2.txt > cleaned_log.txt
  261  sed 's/user/student/gI' lab_2.txt > modified_log.txt
  262  sed 's/^/Log Entry: /' lab_2.txt > labeled_log.txt
  263  sed '/successfully/!d' lab_2.txt > success_log.txt
  264  awk '$3=="ERROR" {$3=""; print}' lab_2.txt
  265  awk 'END {print "Total lines:", NR}' lab_2.txt
  266  awk '/Executing command/ {printf "%s %s ", $1, $2; for(i=4;i<=NF;i++) printf "%s ", $i; print ""}
' lab_2.txt
  267  awk '$3=="INFO" {count++} END {print count}' lab_2.txt
  268  awk '/memory/ {print $1, $2, $0}' lab_2.txt
  269  awk '/access/ {print NR, $0}' lab_2.txt
  270  mkdir -p lab2/dir1/dir2/dir3 lab2/dir1/dir4
  271  mv lab_2.txt updated_log.txt warnings.txt cleaned_log.txt modified_log.txt labeled_log.txt succes
s_log.txt lab2/dir1/dir2/dir3/
  272  tar -czvf lab2.tar.gz lab2/
  273  tar -tzvf lab2.tar.gz
  274  rm -r lab2
  275  tar -xzvf lab2.tar.gz
  276  history
```